



PHYSICS

Stage 2

WACE Examination 2011

Marking Key

Marking keys are an explicit statement about what the examiner expects of candidates when they respond to a question. They are essential to fair assessment because their proper construction underpins reliability and validity.

When examiners design an examination, they develop provisional marking keys that can be reviewed at a marking key ratification meeting and modified as necessary in the light of candidate responses.

Question 1

(3 marks)

Medical workers are instructed to reduce their health risk from exposure to radioisotopes by following the procedures listed below. Explain how each procedure protects medical workers.

Limit time of exposure

Maximise distance from radiation sources

Wear protective clothing and use protective shielding devices

Description	Marks
Reduces total absorbed dose	1
Reduces concentration of radiation (inverse square)	1
Blocks radiation from penetrating to living tissue	1
	Total 3

Question 2

(3 marks)

Answer **TRUE** or **FALSE** to each of the following.

You place a mercury thermometer and an alcohol thermometer into the same beaker of warm liquid at the same time. When the mercury and alcohol bars reach steady points:

the mercury has the same temperature as the warm liquid.

the alcohol has the same temperature as the mercury.

the mercury and alcohol have both absorbed the same amount of heat from the warm liquid.

Description	Marks
True	1
True	1
False	1
	Total 3

Question 3

(6 marks)

A sudden and strong thunderstorm caused a 40.0 kg branch to break off a tree and fall from 9.00 m above the ground. The branch hit the roof of a house under the tree. Assume the branch was in free fall and the average height of the roof was 3.50 m above the ground.

- (a) Calculate the speed of the branch when it hit the roof. (3 marks)

Description	Marks
$v^2 = u^2 + 2(as)$	1
$= 0^2 + 2 \times 9.80 \times 5.50$	1
$v = 10.4 \text{ m s}^{-1}$	1
	Total 3

- (b) The roof was strong enough to withstand the impact and the branch settled on the roof. If the impact contact time was 0.400 seconds, calculate the magnitude of the average force exerted on the roof during the impact. (3 marks)

Description	Marks
$Ft = \Delta p = m(v - u)$	1
$F = \frac{40.0(10.4 - 0)}{0.400}$	1
$F = 1040 \text{ N}$	1
	Total 3

Question 4

(4 marks)

A plastic iceblock tray containing 250 mL of water, initially at 18°C, is placed in the freezer compartment of a refrigerator. How much heat must be lost from this water when it has all become ice at 0°C?

Description	Marks
$Q = mc\Delta T + mL_f$	1
$= 0.25 \times 4.18 \times 10^3 \times 18 + 0.25 \times 3.34 \times 10^5$	2
$= 1.02 \times 10^5 \text{ J}$	1
If $mc\Delta T$ or mL_f omitted max 2	Total 4

Question 5

(3 marks)

A solar hot water system collects solar energy at the rate of 800 J s^{-1} . It needs $1.68 \times 10^7 \text{ J}$ of energy to heat the water it contains. Calculate how many hours it would take to heat this water. Ignore any heat lost to the surroundings.

Description	Marks
$P = \frac{\Delta E}{t}$	1
$t = \frac{\Delta E}{P} = \frac{1.68 \times 10^7}{800} = 2.10 \times 10^4 \text{ s}$	1
$= 5.83 \text{ h}$	1
	Total 3

Question 6

(3 marks)

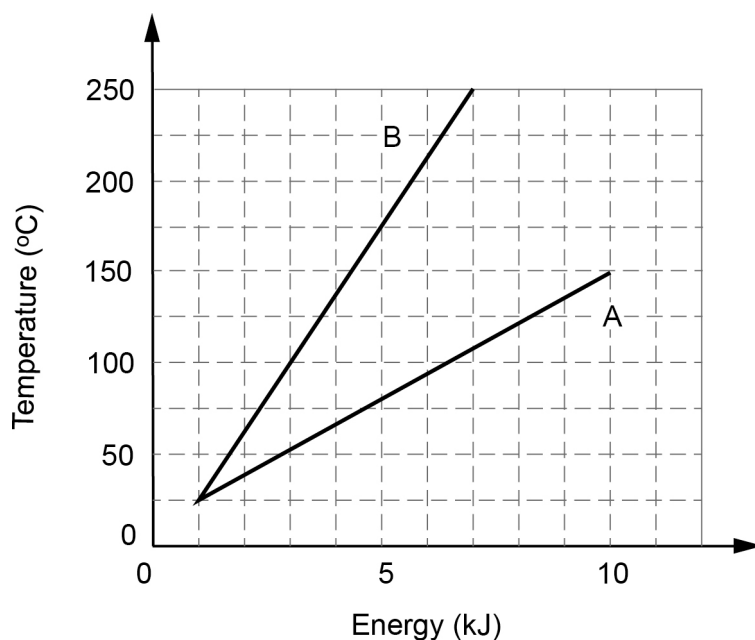
When we go to bed on a winter's night we usually cover ourselves with a blanket, doona or quilt to keep warm. Explain briefly how this keeps us warm by referring to the three methods of heat transfer listed below.

Description	Marks
Conduction: covering is a poor conductor, absorbing little heat from the body and losing little to the air.	1
Radiation: covering absorbs heat being radiated from the body. This heats the covering and the air trapped in it so its temperature rises.	1
Convection: covering traps convection currents keeping warm air in contact with the body.	1
	Total 3

Question 7

(6 marks)

Two objects, A and B, have the same sizes but different masses. The specific heat capacity of each is $234 \text{ J kg}^{-1} \text{ K}^{-1}$. A and B each receive the same amount of energy from an external source. The graph shows the result.



- (a) Which object has the larger mass? Using the correct mathematical formula, explain your reasoning for this choice. (3 marks)

Description	Marks
Object A;	1
Because $Q = mc(T_f - T_i)$, and $m = Q / c(T_f - T_i)$	1
For same material, the greater mass needs more heat energy to raise its temperature.	1
	Total 3

- (b) From the data in the graph, determine the mass of object A. (3 marks)

Description	Marks
$m_A = Q / c(T_f - T_i)$	1
$m_A = 9000 / 234 (150 - 25) = 0.308 \text{ kg or } 308 \text{ g}$	1–2
	Total 3

Question 8 (5 marks)

A beam of electrons in the electron gun of a cathode ray tube is accelerated from rest through a potential difference of 80 kV.

- (a) How much energy in joules does each of these electrons gain in being accelerated in this field? (2 marks)

Description	Marks
$E = qV = 1.60 \times 10^{-19} \times 8.0 \times 10^4$	1
$= 1.28 \times 10^{-14} \text{ J}$	1
	Total 2

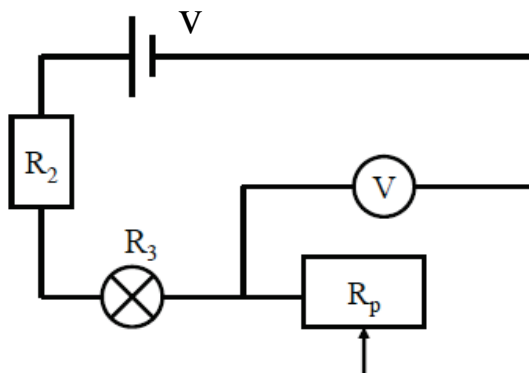
- (b) If the energy gained by each electron is all kinetic, what velocity will an electron have after undergoing this acceleration? (3 marks)

Description	Marks
$0.5mv^2 = 1.28 \times 10^{-14}$	1
$v^2 = 1.28 \times 10^{-14} / (0.5 \times 9.11 \times 10^{-31})$ $= 2.81 \times 10^{16}$	1
$v = 1.68 \times 10^8 \text{ m s}^{-1}$	1
	Total 3

Question 9

(3 marks)

A circuit diagram is shown below. The voltage V of the power supply is constant. R_p is a potentiometer (variable resistor). R_2 has a constant value.



When the resistance of the potentiometer (R_p) increases (the resistance of R_2 and globe R_3 remaining the same), which of the following happens (ignore the resistance of the globe)?

- (i) The reading of the voltmeter increases and the globe is brighter.
- (ii) The reading of the voltmeter decreases and the globe is darker.
- (iii) The reading of the voltmeter increases and the globe is darker.
- (iv) The reading of the voltmeter decreases and the globe is brighter.

Answer: _____

Give reasons for your choice by completing the statements below:

The voltmeter reading _____

because _____

The globe is _____

because _____

Description	Marks
(iii)	1
the reading of the voltmeter increases because R_p increases	1
globe is darker – as R_p increases and V is constant $\frac{V}{R}$ Current must decrease (alternatively power must decrease)	1
Total 3	

Question 10

(6 marks)

A power pack for a laptop computer delivers 19.5 V with a current of 2.05 A. It is connected to the computer to recharge the battery for 2.50 hours.

- (a) How much charge flows from the power pack to the battery in that time? (3 marks)

Description	Marks
$t = 2.50 \times 60 \times 60 \text{ s} = 9.00 \times 10^3 \text{ s}$	1
$Q = It = 2.05 \times 9.00 \times 10^3$	1
$= 1.85 \times 10^4 \text{ C}$	1
	Total 3

- (b) How much work is done in moving this charge? (3 marks)

Description	Marks
Work = qV	1
$= 1.85 \times 10^4 \times 19.5$	1
$= 3.61 \times 10^5 \text{ J}$	1
	Total 3

Question 11

(3 marks)

Many multi-outlet power boards are rated for a maximum current of 7.50 A. Why should we **not** connect a portable electric heater that is rated at 240 V, 2.4 kW to such a board? Explain, showing relevant calculations.

Description	Marks
Current drawn by heater = $\frac{P}{V} = \frac{2400}{240}$	1
$= 10 \text{ A}$	1
This exceeds the safe design current of the power board.	1
	Total 3

Question 12

(3 marks)

Select one electrical safety device you have studied this year and explain how it works.

Description	Marks
Naming fuse, circuit breaker, RCD	1
Detail on correct function – stops current flow when short circuit (fuse, cct breaker) or imbalance in active/neutral current is detected (RCD)	1–2
	Total 3

Question 13

(7 marks)

Calculate the average binding energy per nucleon (in joules) for carbon-14.

Use the mass of carbon-14 = 2.32478×10^{-26} kg.

Use the **Formulae and Constants sheet** for the masses of the neutron and the proton.

Description	Marks
Theoretical mass = $8 \times 1.68 \times 10^{-27} + 6 \times 1.67 \times 10^{-27}$ = 2.346×10^{-26} kg	1–2
Mass defect = theoretical – actual = $2.346 \times 10^{-26} - 2.325 \times 10^{-26}$ = 2.10×10^{-28}	1–2
Total binding energy = $\Delta mc^2 = 2.10 \times 10^{-28} \times (3.00 \times 10^8)^2$ = 1.89×10^{-11} J	1–2
Av binding energy/nucleon = total E_b /total nucleons = $1.89 \times 10^{-11}/14$ = 1.35×10^{-12} J	1
	Total 7

Question 14

(3 marks)

Use the mathematical relationship (formula) for momentum to explain why a slow-moving train will generally have more momentum than a high-velocity small-calibre (diameter) rifle bullet. (Calculations are not required.)

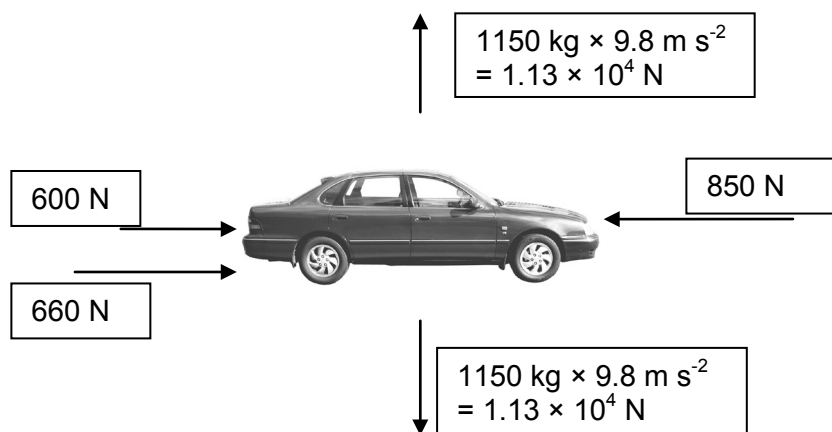
Description	Marks
$p = mv$	1
The train's mass is billions of times bigger than the mass of the bullet. The bullet's velocity is only hundreds of times bigger than the train's That is, ratio $\frac{m_{\text{train}}}{m_{\text{bullet}}} \gg \gg \gg \gg \gg$ ratio $\frac{v_{\text{train}}}{v_{\text{bullet}}}$	1–2
	Total 3

Question 15

(6 marks)

Sam and Henry run out of petrol not far from a service station and decide to push their car to the service station to refuel it. Sam is able to exert a force of 600 N and Henry a force of 660 N. The mass of the car is 1150 kg and there is a frictional force of 850 N. Both boys push from the rear of the car in the same direction.

- (a) Using the picture below, show all of the forces, with their magnitudes, acting on the car under these conditions. (3 marks)



Description	Marks
1 mark subtracted for each error on force diagram to a max of 3	1–3
	Total 3

- (b) What is the magnitude of the resultant force acting on the car under these conditions? (3 marks)

Description	Marks
$600 + 660 = 1260$	1
$F_r^2 = (1260 - 850)$	1
$= 410 \text{ N}$	1
	Total 3

End of Section One

Section Two: Problem-solving

(80 Marks)

Question 16

(8 marks)

An isotope of thorium-232 undergoes radioactive decay via a number of α and β decays. The overall decay equation can be written as;



- (a) Calculate (showing all working) the values of n and m .

(4 marks)

Description	Marks
α is ${}_2^4\text{H}$ and β is ${}_{-1}^0\text{e}$	1–4
$220 + 4n + 0 = 232$ (eq 1)	
$4n = 232 - 220 = 12$; so $n = 3$	
$86 + 2n + (-1)m = 90$ (eq 2)	
Substitute $n = 3$ into (eq 2) $86 + 2 \times 3 - m = 90$ $m = 86 + 6 - 90 = 2$ $m = 2$	
	Total 4

Thorium-232 is the most commonly occurring isotope of thorium and is a solid with a half-life of around 14.05 billion years. It occurs naturally and is in plentiful supply on the Earth's surface. Radon-220 is a gas at room temperature. It is also radioactive, with a half-life of approximately 55 seconds. It decays to polonium-216, with the release of an alpha particle.

- (b) Explain what is meant by 'the half-life of radon-220 is 55 seconds'.

(2 marks)

Description	Marks
It takes 55 s for half of the radon atoms initially present to undergo radioactive decay	1–2
	Total 2

- (c) Even though radon gas has a very short half-life and is not abundant, it is more dangerous to humans than thorium. Explain why.

(2 marks)

Description	Marks
Gas – easily taken into lungs – internal dose more dangerous.	1
Either one of: Decays to solid polonium which remains in the lungs, itself emitting radiation. or Alpha emitter is highly ionising and the most dangerous form of radiation to body cells.	1
	Total 2

Question 17

(14 marks)

Adam returns from Singapore with a lamp only to find that the plug on the lamp does not fit the power outlets in Australia. The lamp consists of a plug, a switch and a globe. The tag on the lamp says it is designed for a 110 V power supply. The Australian power supply voltage is 240 V.

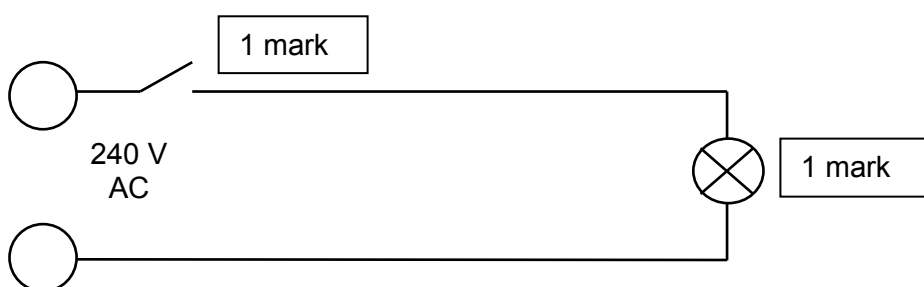


Lamp



Lamp plug

- (a) Using appropriate circuit symbols complete the circuit diagram for this lamp when connected to the 240V AC power supply. (3 marks)



The globe in the lamp is marked '110 V, 0.60 A'.

Description	Marks
Each component shown	1–2
Switch and globe in series	1
	Total 3

- (b) What is the power rating of this globe when it is used in Singapore? (2 marks)

Description	Marks
$P = VI = 110 \times 0.6$	1
$= 66 \text{ W}$	1
	Total 2

- (c) What is the theoretical resistance of this globe under these conditions? (2 marks)

Description	Marks
$R = V/I = 110/0.6$	1
$= 183 \Omega$	1
	Total 2

- (d) Adam buys a plug suitable for Australian power outlets, cuts the Singaporean plug from the wires and connects the Australian plug. Assuming that he has wired it correctly, is the lamp now safe to use in Australia? Explain. (2 marks)

Description	Marks
No. If R is constant then higher V will draw more than double the current which will probably exceed the current rating of the wiring.	1–2
Or	
Yes. If R is increased, as in (e) below then current ($I = \frac{V}{R}$) will be less so wiring is safe.	1–2
	Total 2

- (e) The original 110 V globe is changed to an Australian-made globe with a power rating of 75 W. What is the theoretical resistance of the new globe when it is connected to the Perth electricity supply? (3 marks)

Description	Marks
$R = \frac{V^2}{P}$	1
$= \frac{240^2}{75}$	1
$= 768 \Omega$	1
	Total 3

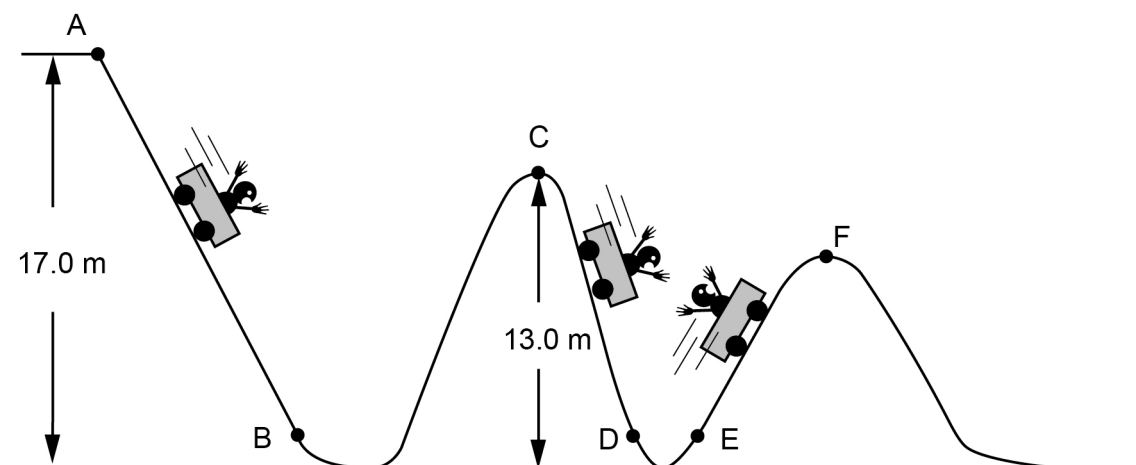
- (f) In (c) you were asked to calculate the theoretical resistance. This is because an operating globe is a non-ohmic conductor. Explain what is meant by the term 'non-ohmic conductor'. (2 marks)

Description	Marks
V not proportional to I – Does not obey Ohm's Law	1–2
	Total 2

Question 18

(14 marks)

A roller-coaster is 17.0 m high at its highest point, the release point A. The diagram below is a simple representation of the first part of the ride.



- (a) The car is released from a stationary position at A and has no independent locomotion. Indicate the direction of the car's acceleration in each of the following regions by circling the correct answer. (3 marks)

- (i) A to B up slope down slope
- (ii) C to D up slope down slope
- (iii) E to F up slope down slope

Description	Marks
down slope	1
down slope	1
down slope	1
	Total 3

- (b) If the length of track between A and B is 21.0 m and the car is released from rest at A and reaches a velocity of 18.5 m s^{-1} at B, what is the magnitude of the acceleration it experiences between A and B? (4 marks)

Description	Marks
$a = (v^2 - u^2)/2s$	1
$= 18.5^2/2 \times 21$	1
$a = 8.15 \text{ m s}^{-2}$	1
Correct units	1
	Total 4

- (c) If the car **just** makes it to point C, what proportion (fraction) of the energy has been lost as heat? (3 marks)

Description	Marks
E_{GP} proportional to h	1
Change in $h = 17 - 13 = 4$ m	1
percentage lost = $\frac{4}{17} =$ (accept 23.5% or 0.235)	1
	Total 3

- (d) In relation to this energy lost as heat, how is it generated and where does it go? (2 marks)

Description	Marks
Friction generates heat $\rightarrow$ main cause of energy loss	1
Very small amount to sound, light (sparks?) and wear (deformation) Heat lost to tracks and air.	1
	Total 2

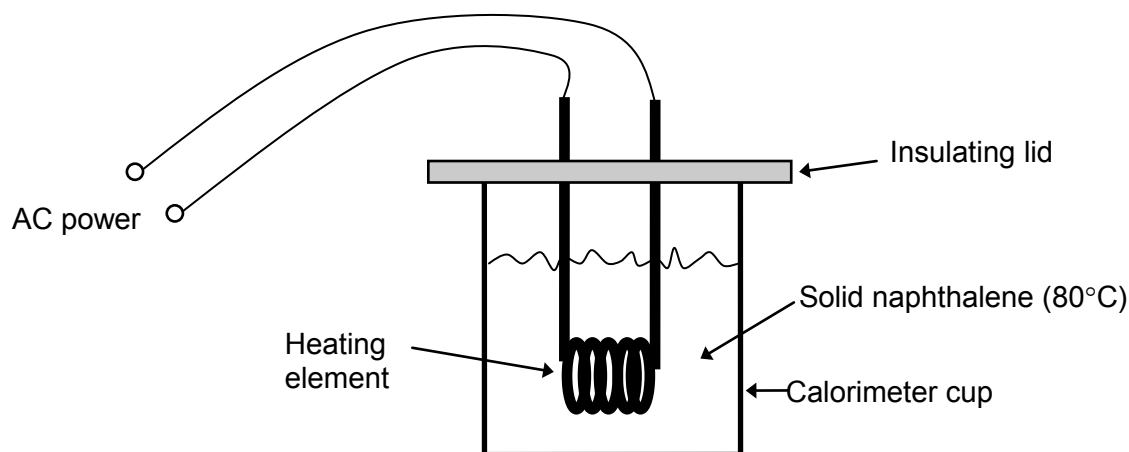
- (e) Will the velocity of the car be greater at point B or point D? Give a reason for your answer. (2 marks)

Description	Marks
Will be greatest at B	1
Earlier in the ride = less heat loss therefore same height E_K is greater at B	1
	Total 2

Question 19

(17 marks)

Hayley investigated the accuracy of the stated power rating on a heating element that was marked '100 W'. She placed solid naphthalene at its melting point of 80.0°C in a calorimeter cup at 80.0°C together with the heating element, and timed how long it took to melt varying amounts of naphthalene.



She collected the following results:

Mass of naphthalene (kg)	Time to melt (s)
0.150	288
0.250	480
0.350	670
0.450	865
0.550	1055

$$L_f(\text{Naphthalene}) = 1.48 \times 10^5 \text{ J kg}^{-1}$$

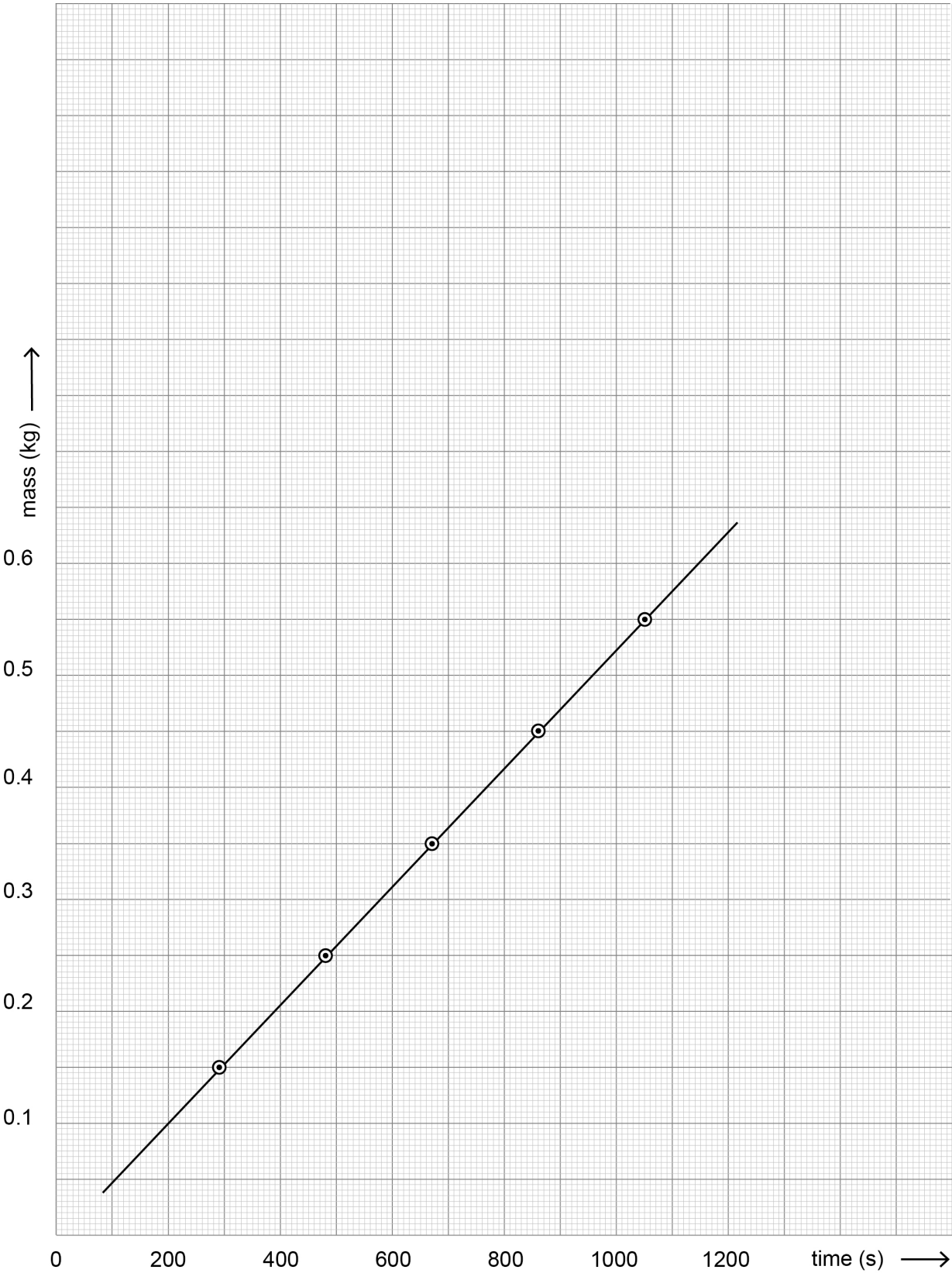
She then used the following relationship to decide how to plot the data:

$$P = \frac{Q}{t} = \frac{mL_f}{t}$$

She used the gradient of the line from her plot to calculate the actual power output of the heating element.

- (a) Draw a clearly-labelled graph on the graph paper provided on page 19. Plot time on the x -axis and mass on the y -axis. (5 marks)

Description	Marks
Correct scale on axes, correct labels and orientation of axes follows instructions	1–3
Data plotted accurately	1
Line of best fit drawn accurately	1
	Total 5



- (b) From the graph, obtain the gradient of the line you have plotted. Show all workings and include the units in your answer. (5 marks)

Description	Marks
Gradient is derived from the graph. ($\frac{\text{rise}}{\text{run}}$)	1
Substitute values	2
gradient = $5.24 \times 10^{-4} \text{ kg s}^{-1}$	1
Correct units	1
	Total 5

- (c) Use the gradient of the graph and the equation to calculate the power output of the heating element. (3 marks)

Description	Marks
$P = \frac{mL}{t} = \text{gradient} \times L$	1
$= 5.24 \times 10^{-4} \times 1.48 \times 10^5$	1
$= 77.6 \text{ W}$	1
	Total 3

- (d) From the graph, determine the time taken to melt 0.300 kg of naphthalene. (2 marks)

Description	Marks
560 - 580s	2
Or 550 - 559 or 581 - 590	1
	Total 2

- (e) Would you expect the calculated power rating to be higher or lower than the true power rating (not the rating printed on the heating element)? Explain your answer. (2 marks)

Description	Marks
As Q can only be lost from the system, input Q needs to be much larger than measured	1
Therefore power rating is really higher	1
	Total 2

Question 20

(10 marks)

A flea's jump is one of the most impressive examples of acceleration in the animal kingdom. By pushing its legs against the ground, the flea can attain an initial upward velocity of 1.00 m s^{-1} in 10.0 milliseconds.

Calculate:

- (a) the flea's average acceleration over this time (that is, while leaving the ground).

(3 marks)

Description	Marks
$a = \frac{(v - u)}{t}$	1
$a = \frac{1}{0.01}$	1
$a = 100 \text{ m s}^{-2}$	1
	Total 3

- (b) the force acting on the flea during this time if it has a mass of 2.00 mg.

(4 marks)

Description	Marks
$F = ma$	1
$F = 2 \times 10^{-6} \times 100$	1–2
$F = 2 \times 10^{-4} \text{ N}$	1
	Total 4

- (c) the maximum height that the flea can reach if its initial velocity is vertically upward.

Hint: once the flea leaves the ground, it is affected only by gravity.

(3 marks)

Description	Marks
$s = \frac{(v^2 - u^2)}{2a}$	1
$s = \frac{-1}{-19.6}$	1
$s = 0.051 \text{ m}$	1
	Total 3

Question 21

(10 marks)

Technetium-99m (Tc-99m) has a half-life of 6 hours and emits only gamma rays. It is used as a radioactive tracer to diagnose illness, which means it is swallowed and collects in the part of the body that needs to be seen. A detector is then used to form an image of that part of the body.

- (a) Select a characteristic of Tc-99m that makes it a good choice for use as a radioactive tracer and explain the benefit of this characteristic. (2 marks)

Description	Marks
Gamma emitter	1
radiation needs to penetrate to be detected outside the body	1
OR Short half-life – decays to a safe level relatively quickly	
	Total 2

- (b) The initial activity of a sample used as a radioactive tracer is 320 Bq. While no activity is safe, it is considered that an activity of 5.00 Bq gives a significantly-reduced risk. How long after swallowing can a patient expect to wait to achieve a significantly-reduced risk? (3 marks)

Description	Marks
320 160 80 40 20 10 5	1
n = 6 half lives	1
t = 6×6 = 36 hours	1
	Total 3

- (c) The energy released to a particular body organ of mass 3.00 kg that Tc-99m collects in is 3.10×10^{-2} J in one minute. What is the absorbed dose in this time? (3 marks)

Description	Marks
Absorbed dose = $\frac{\text{energy}}{\text{mass}}$	1
= $\frac{0.031}{3}$	1
= 1.03×10^{-2} Gy	1
	Total 3

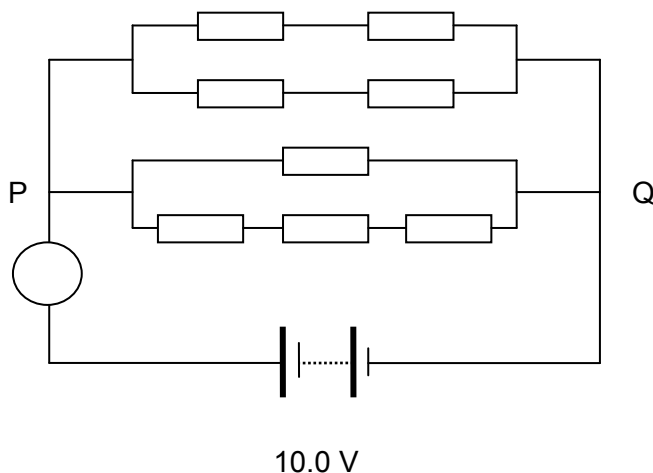
- (d) If a different radioisotope that emitted only alpha radiation was used by mistake to produce the same absorbed dose, it would have a much higher dose equivalent. Give an explanation for this difference. (2 marks)

Description	Marks
Explain dose equiv = abs dose × QF	1
QF of alpha is 20× greater	1
	Total 2

Question 22

(7 marks)

Eight equal value resistances are connected between P and Q. The resistance of each of these resistors is 10.0Ω .



- (a) Calculate the total resistance of the circuit, excluding the meter and the power source. (5 marks)

Description	Marks
$R_{PQ1} = R + R = 2R = 20 \Omega$	1
$R_{PQ2} = R + R = 2R = 20 \Omega$	1
$R_{PQ3} = R = 10 \Omega$	1
$R_{PQ4} = R + R + R = 3R = 30 \Omega$	1
$R_{total} = R_{PQ1} // R_{PQ2} // R_{PQ3} // R_{PQ4}$	1
$1/R_t = 1/2R + 1/2R + 1/R + 1/3R = 14/6R \Rightarrow R_t = 6R/14 \Rightarrow R_t = 4.29 \Omega$	1
Total 5	

- (b) The circle represents a meter. What is the reading on this meter (include the unit)? (2 marks)

Description	Marks
$I = V/R_t = 10 / 4.29 = 2.33 \text{ A}$	1–2
Total 2	

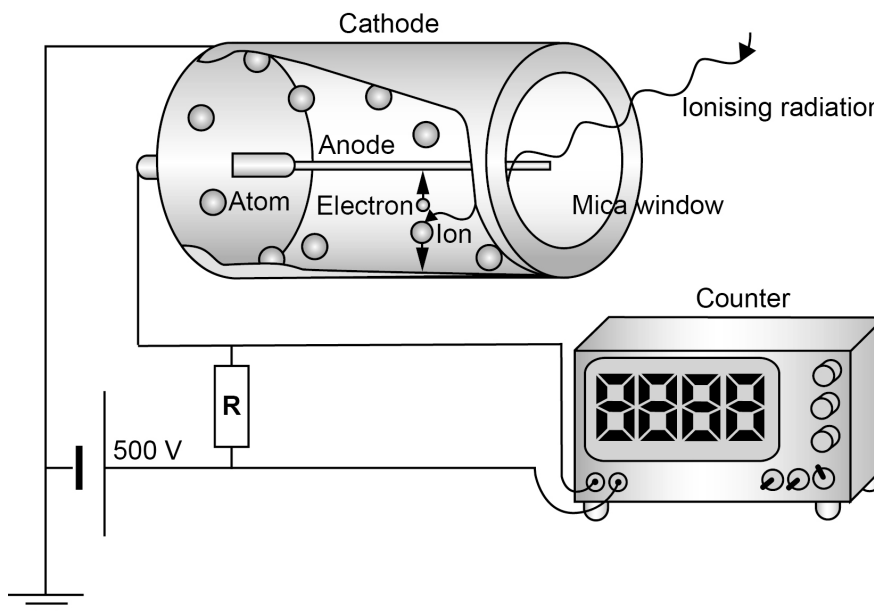
End of Section Two

Section Three: Comprehension

10% (16 Marks)

Question 23

A Geiger–Müller tube (or GM tube) is the sensing element of a Geiger counter instrument that can detect a single particle of ionising radiation.



A GM tube consists of a tube filled with a low-pressure inert gas such as helium, neon or argon. The tube contains electrodes, between which there is a potential difference of several hundred volts, but no current flowing. The walls of the tube are either entirely metal or have their inside surface coated with a conductor to form the cathode while the anode is a wire passing up the centre of the tube.

When ionising radiation passes through the tube, some of the gas molecules are ionised, creating positively charged ions, and electrons. The strong electric field created by the tube's electrodes accelerates the ions towards the cathode and the electrons towards the anode. The ion pairs gain sufficient energy to ionise further gas molecules through collisions on the way, creating an avalanche of charged particles. This results in a short, intense pulse of current that passes (or cascades) from the negative electrode to the positive electrode and is measured or counted.

The number of pulses per second measures the intensity of the radiation present. Some Geiger counters display an exposure rate, but this does not relate easily to a dose rate as the instrument does not discriminate between radiations of different energies.

The usual form of tube is an end-window tube. This type is so-named because the tube has a window at one end through which ionising radiation can easily penetrate. The other end normally has the electrical connectors. There are two types of end-window tubes: the glass-window type and the mica (which can be split into very thin sheets) window type. The glass-window type will not detect alpha radiation since it is unable to penetrate the glass, but is usually cheaper and will usually detect beta radiation and gamma rays. The mica window type will detect alpha radiation but is more fragile.

- (a) Explain how the ionising radiation causes the inert gas inside the GM tube to become charged. (2 marks)

Description	Marks
Radiation ionises the gas atoms by removing electrons producing positive ions.	1–2
	Total 2

- (b) On the circuit diagram on Page 24, draw an arrow to indicate the direction of conventional current flow. (1 mark)

Description	Marks
Either side of power source on the circuit line away from the positive or towards the negative.	1
	Total 1

- (c) With reference to the properties of alpha particles, infer why a mica window will allow the detection of the alpha particles while a glass window will not. (2 marks)

Description	Marks
Alpha particles are easily stopped when they pass through matter.	1
Mica is thin.	1
	Total 2

- (d) During one experiment, the detector indicates 648 counts over a period of 3 minutes when located 10 cm from a source.

- (i) Before the activity of the source can be calculated accurately, the background count is needed. What is meant by the 'background count' and how is it produced? (2 marks)

Description	Marks
Naturally occurring radiation in the environment	1
Comes from cosmic, sources, rocks and minerals	1
	Total 2

- (ii) If the background count is 14 counts per minute, calculate the source's activity. (3 marks)

Description	Marks
Counts per min = $648/3 = 216$	1
Counts per min due to source = $216 - 14 = 202$	1
Activity = $202/60 \text{ Bq} = 3.37 \text{ Bq}$	1
	Total 3

- (iii) As the GM tube moves closer to the source, what would you expect the counter to indicate? (2 marks)

Circle the correct answer and give reasons for your choice:

more activity

less activity

same activity

Description	Marks
More activity	1
Less radiation is stopped by the air between the source and detector as the source gets closer to the detector (or inverse square law).	1
	Total 2

- (e) During another experiment, 5.6×10^{12} electrons moved through the circuit over 0.02 seconds when detecting a radioactive source. Calculate the current in the circuit for this event. (2 marks)

Description	Marks
$q = 5.60 \times 10^{12} \times 1.60 \times 10^{-19} \text{ C}$	1
$I = q/t = 8.96 \times 10^{-7} / 0.02 = 4.48 \times 10^{-5} \text{ A}$	1
	Total 2

- (f) Give two reasons why this device can determine a decay rate, but not a dose rate. (2 marks)

Description	Marks
Machine doesn't discriminate between radiations of different energies.	1
Machine doesn't discriminate between different types of radiation – no quality factor determinations.	1
	Total 2

End of questions

ACKNOWLEDGEMENTS

Section Three: Question 23

Adapted from: knott, T. (2005, October 18). *Geiger–Müller tube* [Drawing]. Retrieved August 25, 2010, from: http://en.wikipedia.org/wiki/Geiger-M%C3%BCller_tube. This file is licensed under the Creative Commons Attribution-Share Alike 3.0 Unported license.